Large Sample Theory

Lecture 3

Our maintained assumptions so far have been:

A1 The model is linear and given by $y_i = x_i'\beta + u_i$.

A2 The rank of the matrix X is p.

A3 The conditional expected value of y is equal to $E(y_i|X) = x_i'\beta$ for all subjects i in the sample of size n.

A4 The conditional variance of y is equal to $V(y|X) = V(X\beta + u|X) = V(u) = \sigma^2 I$, where $\sigma^2 > 0$ and I is the identity matrix.

It is often assumed that $u_i \sim \mathcal{N}(0, 1)$, but this condition is too strong. We can replace it by a weaker condition on the moments of the random variables:

A5 The observations $\{(y_i, x_i); i = 1, 2, ..., n\}$ are *i.i.d.* across i and their fourth moments exist, i.e., $E(z_i^4) < \infty$ for a generic variable z. Also $E(x_i x_i')$ is a non-singular matrix.

We assume that y_i and x_i are jointly distributed and their fourth moments exists (e.g., $E(z^4) < \infty$ for $z = \{x, y\}$). An implication is that the error term u_i does not need to be normally distributed but their fourth moment should exist: $E(u^4) < \infty$.

1 Asymptotics as a tool

Several results obtained in econometrics are based on the normality assumption. This is certainly not a mild assumption and does not hold in practice.

Asymptotics is a tool that helps us to approximate the distribution of estimators or test statistics without assuming a parametric distribution, provided of course that the number of observations n (i.e. sample size) is sufficiently 'large'. Most econometric methods used in applied economics perform well when the sample size is large. Moreover, sometimes is very difficult to obtain the standard error of non-linear estimators in linear models as in the following examples. We use an asymptotic approximation of the distribution of the estimator called 'delta method' (see Lemma 3.9 in Wooldridge 2010 and Section 5 below).

Example 1. Consider the model $y_t = x_t'\beta + u_t$ with t denoting time, and the interest is in estimating the elasticity parameter, $\eta_j = \frac{\partial E(y|x)}{\partial x_j}\frac{x_j}{E(y|x)}$. An estimator is,

$$\hat{\eta}_j = \hat{\beta}_j \frac{\bar{x}_j}{\hat{y}},$$

but the standard error of $\hat{\eta}_j$ is difficult to obtain because $\hat{y}$ and $\bar{x}_j$ are random variables. We can apply the delta method and obtain the standard error of $\hat{\eta}_j$. (We need to apply chain rule since $\beta_j \frac{x_j}{x'\beta}$).

Example 2. Consider a dynamic model of the form, $y_t = \gamma y_{t-1} + x_t'\beta + u_t$, with variables in logs. The long run elasticity of a change in x_j is, in this case, $\eta_j = \beta_j/(1-\gamma)$. Again, the estimator is non-linear in parameters,

$$\hat{\eta}_j = \frac{\beta_j}{1-\hat{\gamma}},$$

so the standard error of $\hat{\eta}_j$ can be obtained by applying the delta method.

2 Background

2.1 Large sample results

Consider the sequence of random variables $\{X_n : n = 1, 2, \ldots\}$ and let X be a real valued random variable on a well defined probability space.

Definition 1 (Convergence in probability)**.** We say that X_n converges in probability to X if

$$\mathrm{P}(|X_n - X| > \epsilon) \to 0, \quad \text{as } n \to \infty \tag{1}$$

for any $\epsilon > 0$.

Intuitively, the probability that the difference $|X_n - X|$ takes a large value is very small as the sample size n increases. If $X_n \to \beta$, where β is a constant, we say that X_n is consistent and, often, we denote it by $\text{plim}_{n\to\infty} X_n = \beta$.

Why convergence concepts are important? We need a notion of "closeness" to unknown parameters. Recall that we have an i.i.d. (independently and identically distributed) random sample $X_1, X_2, \ldots$ with realizations $x_1, x_2, \ldots$. The joint distribution can be obtained by multiplying the marginal distribution. Suppose also that we are interested in two parameters, the population mean $E(X) = \mu$ and population variance, $\text{Var}(X) = \sigma^2$. We can estimate them with,

$$\begin{aligned}
\bar{X}_n &= T_x(X_1, \ldots, X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i \\
S^2 &= T_s(X_1, \ldots, X_n) = \frac{1}{n}\sum_{i=1}^{n} (X_i - \bar{X}_n)^2.
\end{aligned}$$

How "close" $\bar{X}_n$ is to μ and S^2 is to σ^2? If the sample size is sufficiently large, we expect $(\bar{X}_n, S^2)$ to be close to (μ, σ^2). Now suppose we have a random experiment and we obtain basic outcomes s from the sample space S. We observe $x_i = X_i(s)$ and we can obtain

$$\bar{x}_n = \bar{X}_n(s). \tag{2}$$

Note that different outcomes will create different real numbers $\{\bar{x}_n = \bar{X}_n(s) : n =, 1, 2 \ldots, \}$. Figure 1 illustrates the performance of $\bar{X}_n(s)$ and $S^2(s)$ for 3 different outcomes s_1, s_2 and s_3 from S.

Example 3. Suppose $Y_1, Y_2, \ldots, Y_n$ is an i.i.d. random sample from a uniform distribution $\mathcal{U}(0, \theta)$, where θ is an unknown parameter. It is natural to estimate θ considering the largest order statistic, i.e. $X_n = \max_{1 \leq i \leq n}\{Y_i\}$. Is this estimator consistent? Note that, for a small constant $\epsilon > 0$, we have that $\{|X_n - \theta| > \epsilon\} = \{X_n - \theta > \epsilon\} \cup \{X_n - \theta < -\epsilon\}$. Therefore,

$$\begin{aligned}
P(|X_n - \theta| > \epsilon) &= P(X_n - \theta > \epsilon) + P(X_n - \theta < -\epsilon) = P(X_n > \theta + \epsilon) + P(X_n < \theta - \epsilon) \\
&= P(X_n < \theta - \epsilon) = P\left(\max_{1 \leq i \leq n}\{Y_i\} < \theta - \epsilon\right) \\
&= P(Y_1 < \theta - \epsilon, Y_2 < \theta - \epsilon, \ldots, Y_n < \theta - \epsilon) = \Pi_{i=1}^{n} P(Y_i < \theta - \epsilon) \\
&= \left(\frac{\theta - \epsilon}{\theta}\right)^n = \left(1 - \frac{\epsilon}{\theta}\right)^n \to 0, \text{ as } n \to \infty.
\end{aligned}$$

Definition 2 (Almost sure convergence)**.** We say that X_n converges with probability 1, or almost surely, to X if

$$\mathrm{P}(\lim_{n \to \infty} X_n = X) = 1 \tag{3}$$

Definition 3 (Convergence in q^{th} mean)**.** We say that X_n converges in q^{th} mean to X if,

$$\lim_{n \to \infty} \mathbb{E}|X_n - X|^q = 0 \tag{4}$$

Definition 4 (Convergence in distribution)**.** We say that X_n converges in distribution to X if

$$\lim_{n \to \infty} F_n(X) = F(X). \tag{5}$$

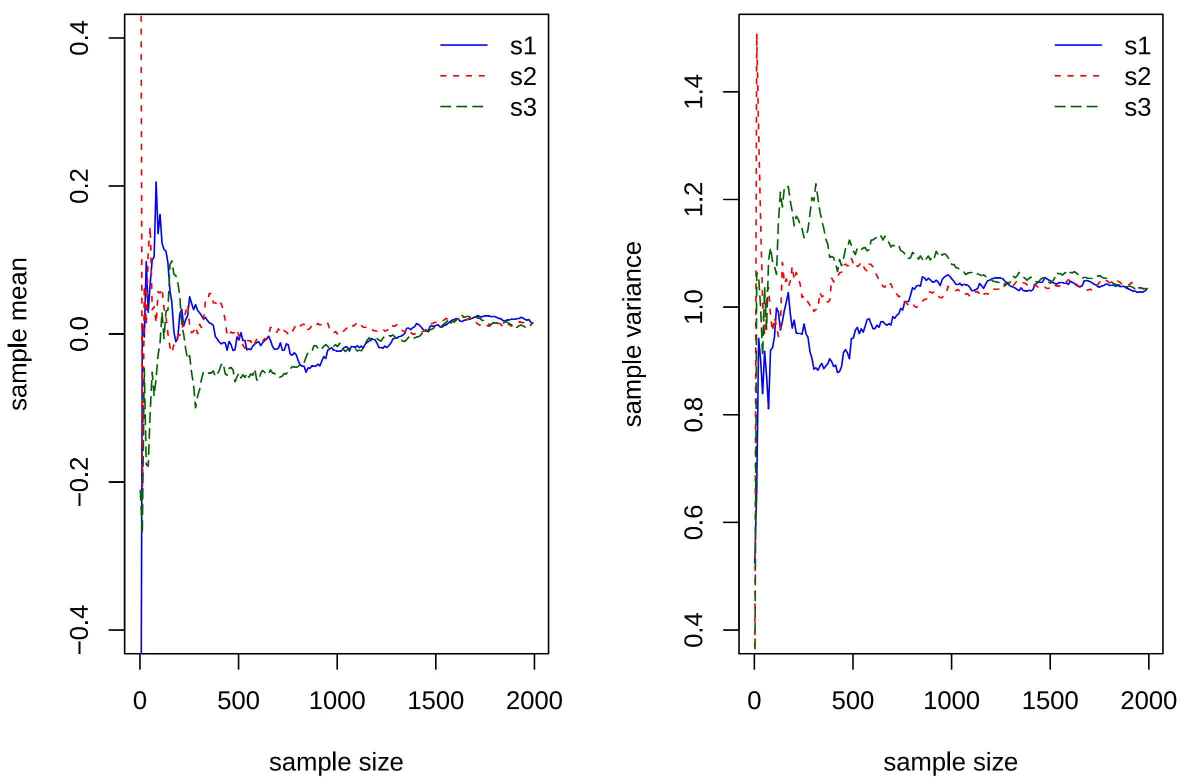

Figure 1: Sample mean and variance as sample size increases defined on a series of basic outcomes s

At each continuity point of F, we denote this as $X_n \xrightarrow{d} X$.

Theorem 1. *If $X_n \xrightarrow{\text{as}} X_1$, then $X_n \xrightarrow{\text{p}} X$, which says that almost sure convergence implies convergence in probability.*

Theorem 2. *If $X_n \xrightarrow{\text{rth}} X$, then $X_n \xrightarrow{\text{p}} X$*

Proof:

$$\mathbb{E}|X_n - X|^r \geq \mathbb{E}\{|X_n - X|^r I(|X_n - X| > \epsilon)\} \geq \epsilon^r \text{P}(|X_n - X| > \epsilon) \tag{6}$$

Therefore,

$$\text{P}(|X_n - X| > \epsilon) \leq \epsilon^{-r} \mathbb{E}|X_n - X|^r \to 0 \text{ as } n \to \infty. \tag{7}$$

■

Corollary 1. *Convergence in probability, $X_n \xrightarrow{\text{p}} X$, implies convergence in distribution, $X_n \xrightarrow{\text{d}} X$.*

Corollary 2. *Convergence in probability, to a constant is equivalent to convergence in distribution to a given constant*

Theorem 3 (Asymptotic equivalence)**.** *Consider a random sample (Y_n, X_n) with $Y_n \xrightarrow{d} y$, and $|Z_n| \xrightarrow{p} 0$, where $Z_n = Y_n - X_n$. It follows that $X_n \xrightarrow{d} y$.*

Proof: For $\epsilon > 0$, we can write $P(X_n < v) = P(Y_n < v + Z_n) = P(Y_n < v + Z_n, Z_n < \epsilon) + P(Y_n < v + Z_n, Z_n \geq \epsilon)$. Note too that $P(Y_n < v + Z_n, Z_n < \epsilon) \leq P(Y_n < v + \epsilon)$. Then it follows that

$$\begin{aligned} P(X_n < v) &\leq P(Y_n < v + \epsilon) + P(Y_n < v + Z_n, z_n \geq \epsilon) \\ &= P(Y_n < v + \epsilon) + [P(Z_n \geq \epsilon) P(Y_n < v + Z_n | z_n \geq \epsilon)] \\ &\leq P(Y_n < v + \epsilon) + P(Z_n \geq \epsilon) \end{aligned}$$

Note that we can similarly write, $P(X_n < v) = P(Y_n < v + Z_n) = P(Y_n < v + Z_n, Z_n > -\epsilon) + P(Y_n < v + Z_n, Z_n \leq -\epsilon)$ which leads to

$$P(X_n < v) \geq P(Y_n < v - \epsilon) + P(Z_n \geq -\epsilon).$$

By the assumption that $|Z_n| \xrightarrow{p} 0$, the last term tends to 0 as $n \to \infty$, so it can be ignored. We can write

$$P(Y_n < v - \epsilon) \leq P(X_n < v) \leq P(Y_n < v + \epsilon).$$

By picking ϵ arbitrarily small, we have that $F_X = F_Y$ as $n \to \infty$. ■

We can interpret this result as showing the asymptotic equivalence when two random variables are very close to each other. As an illustrative example, consider $Y_n = \sqrt{n} \bar{X}_n / S_n$

and $Z_n = \sqrt{n}\bar{X}_n/\sigma$, where σ is an unknown parameter. Dealing with a random denominator S_n is complicated, but we can first show that $Y_n - Z_n \xrightarrow{p} 0$ and then obtain the distribution of Y_n using the result that $Y_n \xrightarrow{d} Z$.

Theorem 4 (Slutsky). *Let $X_n \xrightarrow{\mathrm{d}} x$ and $Y_n \to y$, a real constant. Then,*

1. $X_n + Y_n \xrightarrow{\mathrm{d}} x + y$
2. $X_n Y_n \xrightarrow{\mathrm{d}} yx$
3. $X_n / Y_n \xrightarrow{\mathrm{d}} x/c$ *if* $c \neq 0$

Theorem 5 (Continuous Mapping). *If $X_n \xrightarrow{\mathrm{d}} X$ and g is continuous, $g(X_n) \to g(X)$*

Example 4. If $X_n \xrightarrow{\mathrm{d}} \mathcal{N}(0,1)$, then $X_n^2 \xrightarrow{\mathrm{d}} \chi_1^2$

Example 5. If $(X_n, Y_n) \xrightarrow{\mathrm{d}} \mathcal{N}(0, I_2)$, then $X_n/Y_n \xrightarrow{\mathrm{d}} z$, a Cauchy random variable.

Example 6. When g is not continuous, we may need to be careful. Consider the following example:

$$g(t) = \begin{cases} t-1 & t \leq 0 \\ t+1 & t > 0 \end{cases}$$

Let $X_n = 1/n$, then $X_n \to X = 0$, $g(X_n) \to 1$ but $g(X) = -1$.

2.2 Order relations and rates of convergence

We now examine order relations and rates of convergence. In this section, the terms $O(\cdot)$ ('big oh') and $o(\cdot)$ ('small oh') are used to determine the order of magnitude of deterministic and stochastic sequences.

Consider positive deterministic sequences $\{a_n\}$ and $\{b_n\}$.

Definition 5. $a_n = o(b_n)$ as $n \to \infty$ if $a_n/b_n \to 0$

Perhaps it is good now to have a more comprehensive definition.

Definition 6 (Orders of magnitude). A sequence $\{a_n\}$ is *of order smaller than* n^λ, which is denoted by $a_n = o(n^\lambda)$ or $n^{-\lambda}a_n = o(1)$, if for $n \geq N$, we have that

$$|n^{-\lambda}a_n| < \epsilon,$$

for every $\epsilon > 0$.

Definition 7 (Orders of magnitude)**.** A sequence $\{b_n\}$ is *at most of order* n^λ, which is denoted by $b_n = O(n^\lambda)$ or $n^{-\lambda}b_n = O(1)$, if for $n \geq N$, we have that

$$|n^{-\lambda}b_n| < K,$$

where $K < \infty$ is a very large real number.

Remark 1. a_n and b_n tends to infinity, this says that a_n tends to infinity more slowly than b_n. If a_n and b_n tend to zero, this indicates that a_n tends to zero faster that b_n

Example 7. $1/n^2 = o(1/n)$ as $n \to \infty$, because $(1/n^2)/(1/n) = n/n^2 = 1/n \to 0$

Remark 2. It makes a big difference whether a sequence tends to 0 at rate $1/n^2$ or $1/n$

Suppose

$$a_n = \frac{1}{n} - \frac{2}{n^2} + \frac{4}{n^3} \tag{8}$$

Let's approximate a_n by $b_n = 1/n$. Therefore,

$$a_n = \frac{1}{n} + R_n = \frac{1}{n} + o(1/n) \tag{9}$$

Where $R_n = \frac{-2}{n^2} + \frac{4}{n^3} = o(1/n)$. Or, if we approximate a_n by $1/n - 2/n^2$, we write

$$a_n = \frac{1}{n} - \frac{2}{n^2} + o(1/n^2) \tag{10}$$

Lemma 1. *(a)* $a_n = o(1)$ iff $a_n \to 0$*;*

(b) If $a_n = o(b_n)$ *and* $b_n = o(c_n)$*,then* $a_n = o(c_n)$*;*

(c) For any constant $c \neq 0$*,* $a_n = o(b_n)$ *implies* $ca_n = o(b_n)$*;*

(d) For any sequence of numbers $c_n \neq 0$*,*$a_n = o(b_n)$ *implies* $c_n a_n = o(c_n b_n)$*;*

(e) If $d_n = o(b_n)$ *and* $e_n = o(c_n)$*, then* $d_n e_n = o(b_n c_n)$

Example 8. Let $a_n = \log n$, and $b_n = n^k$. Then

$$a_n = o(b_n) \text{ for every } k > 0 \tag{11}$$

Definition 8. If a_n is of n order smaller than or equal to b_n, we say that

$$a_n = O(b_n) \tag{12}$$

If there is a $\Delta < \infty$, such that $a_n/b_n \leq \Delta$ for sufficiently large n, we say that $a_n = O(b_n)$. Note that a particular $a_n = O(1)$ *iff* the sequence a_n is bounded.

Lemma 2. *If $a_n = O(n^r)$ and $b_m = O(n^s)$, then:*

(a) $a_n b_n = O(n^{r+s})$

(b) $a_n + b_n = O(n^{max\{r,s\}})$

The terms $O_p(\cdot)$ ('big oh') and $o_p(\cdot)$ ('small oh') are for stochastic sequences. Intuitively, we can say that a sequence $\{X_n : n = 1, 2, \ldots\}$ is *of smaller order of magnitude* than n^λ if it converges in probability $X_n/n^\lambda \to 0$ as $n \to \infty$. It is denoted by $X_n = o_p(n^\lambda)$.

Similarly, we can say that a sequence $\{X_n : n = 1, 2, \ldots\}$ is *at most of order* of magnitude n^λ if $P(|X_n/n^\lambda| > M) < \delta$, where $\delta > 0$. It is denoted by $X_n = O_p(n^\lambda)$, and it implies that X_n is bounded in probability.

Now consider sequences $\{X_n\}$ and $\{Y_n\}$ on the probability space $(\Omega, \mathcal{A}, \mathcal{P})$ and any $\epsilon > 0$,

Definition 9. a. If $\exists \Delta < \infty$ such that $\mathrm{P}(|X_n| \geq \Delta |Y_n|) < \epsilon$ for sufficiently large n, we write

$$X_n = O_p(Y_n) \tag{13}$$

If Y_n is non stochastic, we say that X_n is bounded in probability and $X_n = O_p(1)$

b. If $\mathrm{P}(|X_n| \geq \epsilon |Y_n|) \to 0$, we write $X_n = o_p(Y_n)$. We say that it is tending to zero in probability, i.e.,

$$\frac{X_n}{Y_n} \to 0. \tag{14}$$

3 A few LLNs and CLTs

The law of large numbers (LLN) and central limit theorem (CLT) plays a key role in econometrics. For an extensive discussion, please see White (2000) and Davidson (1994). In this section, we will review a few results with particular emphasis on the variations related to the distribution of a sequence of random variables.

Theorem 6 (Khinchine LLN)**.** *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_1) = \mu$. Then,*

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i - \mu \xrightarrow{p} 0. \tag{15}$$

Theorem 7 (Chebyshev LLN)**.** *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_i) = \mu_i$ and $Var(X_i) = \sigma_i^2$. If $n^{-1} \sum_{i=1}^{n} \sigma_i^2 \leq K < \infty$ as $n \to \infty$, then,*

$$\bar{X}_n - \mu_n = \frac{1}{n} \sum_{i=1}^{n} \mu_i \xrightarrow{p} 0. \tag{16}$$

Theorem 8 (Kolmogorov LLN). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_i) = \mu_i$ and $Var(X_i) = \sigma_i^2$. If $\sum_{i=1}^n \sigma_i^2/n^2 < \infty$ as $n \to \infty$, then $\bar{X}_n - \mu_n \longrightarrow 0$ with probability 1 (wp 1).*

The literature on CLT is extensive, so I will only discuss a few results.

Theorem 9 (Lindberg-Levy CLT). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_1) = \mu$ and $Var(X_1) = \sigma^2$. Then,*

$$\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \tag{17}$$

The Liapunov CLT and Linberg-Feller CLT cover the case of heterogeneously distributed random variables, while assuming independence. I introduce the following result which has a broader use in econometrics.

Theorem 10. *Let $\{X_i\}$ be a sequence of independent random variables with $\mathbb{E}(X_i) = \mu_i$, $Var(X_i) = \sigma_i^2$ and $E|X_i - \mu_i|^{2+\delta} < \infty$ for some positive δ. If $\lim_{n\to\infty} n^{-1} \sum_{i=1}^n \sigma_i^2 < \infty$, then,*

$$\sqrt{n}(\bar{X}_n - \bar{\mu}_n)/\bar{\sigma}_n \xrightarrow{d} \mathcal{N}(0, 1). \tag{18}$$

4 The Asymptotic Theory of OLS

4.1 Consistency

Definition 10. A consistent estimator $\hat{\theta}$ is one that converges to θ_0 in probability,

$$\hat{\theta} \to \theta_0, \tag{19}$$

as the sample size n goes to infinity, for all possible true values.

With this definition in mind, we now investigate consistency of OLS in the simplest conceivable setting. Consider a linear regression model of the form,

$$y_i = x_i'\beta + u_i, \tag{20}$$

where $E(u_i|x_i) = E(u_i) = 0$. We can tentatively assume that the error terms u are iid draws from F. The OLS estimator for the unknown parameter β is $\hat{\beta} = (X'X)^{-1}X'y$ and can be written as

$$\begin{aligned} \hat{\beta} - \beta &= (X'X)^{-1}X'u & (21) \\ &= \left(\frac{1}{n}\sum_{i=1}^n x_i x_i'\right)^{-1} \left(\frac{1}{n}\sum_{i=1}^n x_i u_i\right) & (22) \end{aligned}$$

where X is a $n \times p$ matrix and u is a $n \times 1$ vector. If the x's are considered fixed or non-stochastic, we have that the first RHS term,

$$\frac{1}{n}\sum_{i=1}^{n} x_i x_i' \to Q_0 \text{ and } \frac{1}{n}\sum_{i=1}^{n} x_i u_i \to 0.$$

Note that

$$\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} x_i u_i\right) = 0 \tag{23}$$

because $\mathbb{E}(u_i|x_i) = 0$ and the u's are independent. Moreover,

$$\begin{aligned} \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i u_i\right) &= \frac{1}{n^2}\sum_{i=1}^{n} \text{Var}(x_i u_i) && (24)\\ &= \frac{1}{n^2}\sigma^2 \sum_{i=1}^{n} x_i x_i' && (25)\\ &= \frac{\sigma^2}{n}\frac{1}{n}\sum_{i=1}^{n} x_i x_i' \to 0 && (26) \end{aligned}$$

Then,

$$\hat{\beta} \to \beta, \tag{27}$$

follows by a simple application of Chebychev and the Slutsky theorem. (The case of stochastic x's requires the use of SLLN, but the argument is similar).

Remark 3 (Inconsistency 1)**.** Note that if $x_i = i$ in a model with no intercept and $p = 1$, then

$$\hat{\beta} - \beta = \left(\frac{1}{n}\sum_{i=1}^{n} i^2\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} i u_i\right).$$

It is immediately apparent that

$$\frac{1}{n}\sum_{i=1}^{n} i^2,$$

does not converge to a constant, say Q_0 as before, and the inverse does not exist. Note that dividing by n^2 will not work either. However, if

$$n(\hat{\beta} - \beta) = \left(\frac{1}{n^3}\sum_{i=1}^{n} i^2\right)^{-1}\left(\frac{1}{n^2}\sum_{i=1}^{n} i u_i\right),$$

the first term on the right hand side converges to $1/3$ as $n \to \infty$, and the variance of the last term converges to zero.

Remark 4 (Super consistency)**.** The super consistent estimator

$$n(\hat{\beta} - \beta) = O_p(n^{-3/2}),$$

while the standard OLS estimator when x_i is not a deterministic trend is

$$\hat{\beta} - \beta = O_p(n^{-1/2}).$$

Remark 5 (Inconsistency 2)**.** If the main equation omits a variable, say variable v_i, then

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^{n} x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^{n} x_i (v_i + u_i) \right).$$

Following previous steps, we obtain

$$\hat{\beta} - \beta \to Q_0 Q_{xv} \neq 0,$$

where Q_{xv} is the covariance matrix of x_i and v_i and it is non-zero if the variables are correlated.

4.2 Asymptotic Normality

Consider (again) a linear regression model:

$$y_i = x_i' \beta + u_i \tag{28}$$

Recall that $\hat{\beta} - \beta = (X'X)^{-1} X'u$. We can write:

$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &= (X'X/n)^{-1} n^{-1/2} X'u \\ &= \left(\frac{1}{n} \sum_{i=1}^{n} x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} x_i u_i \right) \end{aligned}$$

If we apply Corollary 2, we have that

$$\left(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} x_i u_i \right) \xrightarrow{\text{d}} \mathcal{N}(0, \sigma^2 Q_0) \tag{29}$$

By Slutsky theorem, we have that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{\mathrm{d}} Q_0^{-1}\mathcal{N}(0, \sigma^2 Q_0) = \mathcal{N}(0, \sigma^2 Q_0^{-1}) \tag{30}$$

Note that we applied again the Slutsky theorem. In sum, asymptotic normality follows from (i) convergence in distribution of the score, (ii) convergence in probability of the Hessian, and (iii) the Slutsky theorem.

Remark 6. Key to the results derived before is the fact that,

$$\begin{aligned} S &= E(x_i x_i' u_i^2) = E\{E(x_i x_i' u_i^2 | x_i)\} \\ &= E\{x_i x_i' E(u_i^2 | x_i)\} = \sigma^2 E\{x_i x_i'\} = \sigma^2 Q_0, \end{aligned}$$

and if $E(u_i^2|x_i) \neq \sigma^2$, we cannot simplify the expressions as in the last steps.

5 The Delta Method

For the method, the first result that we need is consistency: if the function g is nice and smooth, then $g(\hat{\theta}) \to g(\theta)$. Assuming that $\hat{\theta}$ is consistent and asymptotically normal with variance Ω, and that $g(\cdot)$ is smooth,

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \rightsquigarrow \mathcal{N}(0, G'\Omega G)$$

where

$$G = \nabla_\theta g(\theta)|_{\theta=\hat{\theta}} = \begin{pmatrix} \partial g(\theta)/\partial \theta_1 \\ \partial g(\theta)/\partial \theta_2 \\ \vdots \\ \partial g(\theta)/\partial \theta_p \end{pmatrix}$$

Why? Because Taylor expansion will claim that for $g(\hat{\theta})$ around $\hat{\theta} = \theta$,

$$g(\hat{\theta}) = g(\theta) + (\hat{\theta} - \theta)\nabla_\theta g(\hat{\theta})|_{\hat{\theta}=\theta} + (\theta^* - \theta)'\nabla_\theta^2 g(\hat{\theta})|_{\hat{\theta}=\theta}(\theta^* - \theta)$$

where $\theta_j^* \in (\hat{\theta}_j, \theta_j)\ \forall j$. Note that if the estimator $\hat{\theta}$ is consistent, then the 2nd order term vanishes asymptotically because

$$\hat{\theta}_i \to \theta_j \Rightarrow \theta_i^* \to \theta_j \quad \forall j$$

Therefore, asymptotically we have that

$$V(g(\hat{\theta})) = (\nabla_\theta g(\hat{\theta})|_{\hat{\theta}=\theta})' V(\hat{\theta}) (\nabla_\theta g(\hat{\theta})|_{\hat{\theta}=\theta})$$

6 Introduction to empirical processes

In modern econometrics, point-wise convergence is sometimes too weak to show that the estimator $\hat{\theta}$ is consistent for θ_0. We use a concept that is related to "functional" convergence or uniform convergence of criterion functions. In this section, we will present an introduction to some results that are frequently used.

6.1 Background and definitions

Consider $X_1, \ldots, X_n$ to be i.i.d. with distribution F defined F_n is a random function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} 1_{(-\infty,x]}(X_i),$$

for $X \in R$. Recall that for a fixed x, $F_n(x) \rightarrow F(X)$ by a LLN and

$$Z_n(x) = \sqrt{n}(F_n(x) - F(x)) \rightarrow Z \sim N(0,1).$$

More general uniform convergence results on the empirical process are:

Theorem 11 (Glivenko - Cantelli (1933))**.**

$$\|F_n - F\|_\infty = sup_x |F_n(x) - F(x)| \rightarrow 0$$

6.1.1 M- and Z- estimators

A popular method for finding estimators is to maximize a criterion function of the type

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} m_\theta(X_i),$$

For known functions $m_\theta : \chi \rightarrow R$. For instance, $m_\theta(x_i) = (x - \theta)^2$, $m_\theta(x_i) = |x - \theta|$ or $m_\theta = -\log P_\theta$. Consider the following definitions:

$$\begin{aligned} \theta_0 &= \arg\max_{\theta \in \Theta} M(\theta) \\ \hat{\theta}_n &= \arg\max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} m_\theta(X_i) = \arg\max_{\theta \in \Theta} M_n(\theta) \end{aligned}$$

Another way of thinking this problem is to consider, finding the solution of,

$$\Psi_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} m'_\theta(X_i) = \frac{1}{n}\sum_{i=1}^{n} \Psi_\theta(X_i) = 0$$

The solution of this estimating equation is $\hat{\theta}$ are called Z-estimators. Note however, that the estimating equations may not need to correspond to a maximization problem.

6.2 Uniform consistency

We now investigate consistency for M estimators. Let $\hat{\theta}_n$ be an estimator of θ. We call it asymptotically consistent if the sequence $\hat{\theta}_n$ convergences in probability for θ for every possible value of θ. (e.g. $\bar{X}$ for the population mean EX).

Let d be a metric space that includes the set of possible parameters. We want to prove that $d(\hat{\theta}_n, \theta_0) \to 0$.

The basic point of the issue of point-wise vs functional convergence, it that although

$$M_n(\theta) \to M(\theta) \quad \text{for every } \theta$$

and

$$\arg\max M_n(\theta) \to \arg\max M(\theta),$$

we cannot ensure convergence of $\hat{\theta}_n$. We need a "functional" convergence of M_n to M, to strength out the point-wise convergence. It is commonly used the notion of uniform convergence of the criterion functions.

Consider an estimator $\hat{\theta}_n$ that nearly maximize M_n:

$$M_n(\hat{\theta}_n) \geq \sup_{\theta\in\Theta} M_n(\theta) - o_p(1),$$

which of course implies that

$$M_n(\hat{\theta}) \geq M_n(\theta_0) - o_p(1)$$

Theorem 12. *: For $\epsilon > 0$,*

1. $\sup |M_n(\theta) - M(\theta)| \to 0$,
2. $\sup_{\theta:d(\theta,\theta_0)\geq\epsilon} M(\theta) < M(\theta_0)$

Then, $\hat{\theta}_n \to \theta_0$.

Proof: Consider that $\hat{\theta}_n$ satisfies $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_p(1)$. By uniform convergence of M_n to M by Assumption 1, we have that $M_n(\hat{\theta}_n) \geq M(\theta_0) - o_p(1)$. Then,

$$M_n(\hat{\theta}_n) - M(\hat{\theta}_n) \geq M(\theta_n) - M(\hat{\theta}_n) + o_p(1)$$

$$\sup_{\theta} |M_n(\hat{\theta} - M(\theta)| \geq M(\theta_0) - M(\hat{\theta}_n) + o_p(1)$$

Therefore,

$$M(\theta_0) - M(\hat{\theta}_n) \leq \sup_{\theta} |M_n(\theta) - M(\theta)| + o_p(1) \to 0$$

For every θ, we have, by Assumption 2, that

$$M(\theta) < M(\theta_0) - \eta.$$

Thus, the event $\{d(\hat{\theta}_n, \theta) \geq \epsilon\}$ is contained in the event $\{M(\hat{\theta}_n) < M(\theta_0) - \eta\}$. We can write,

$$P(d(\hat{\theta}_n, \theta) \geq \epsilon) \leq P(M(\hat{\theta}_n) - M(\theta_0) > eta) \to 0,$$

as $n \to \infty$. ■

Remark 7. Uniform convergence of $M_n(\theta)$ required by the stochastic condition is equivalent to the set of functions $\{m_\theta : \theta \in \Theta\}$ being Glivenko, Cantelli. A simple set of sufficient conditions is that Θ be compact, $m_\theta(x)$ be continuous for each x, and the function is dominated by an integrable function.

6.3 Asymptotic Normality

Without lost of generality, let now consider the Z estimators instead of the M estimators. Let a simple $X_1 \cdots X_n$ from a distribution P, and let the "true" criterion function be of the form,

$$\begin{aligned} \Psi_n(\theta) &= \frac{1}{n} \sum_{i=1}^{n} \psi_\theta(X_i) \\ \Psi(\theta) &= E\psi_\theta. \end{aligned}$$

Assume that $\hat{\theta}_n$ is a zero of Ψ_n and converges in probability to a zero θ_0 in Ψ. We expand $\Psi_n(\hat{\theta}_n)$ using Taylor series around θ_0:

$$0 = \Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + (\hat{\theta}_n - \theta_0)\Psi'_n(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)^2 \Psi''_n(\theta^*_n),$$

where $\theta_n^* \in (\theta_0, \hat{\theta}_n)$. Thus,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{-\sqrt{n}\Psi_n(\theta_0)}{\Psi_n'(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)\Psi_n''(\theta_n^*)}. \tag{31}$$

Assuming that $V_\psi = E\psi_{\theta_0}^2$ is finite and considering that $E\psi_{\theta_0} = 0$,

$$-\sqrt{n}\Psi_n(\theta_0) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\theta_0}(X_i) \rightsquigarrow N(0, V_\psi).$$

We now consider the denominator of equation (31). The first term $\Psi_n'(\theta_0) \to E\psi_{\theta_0}'$ by LLN. The second term is

$$o_p(1) \times \psi_n''(\theta_n^*) = o_p(1) \times O_p(1) = o_p(1),$$

because $\hat{\theta}_n - \theta_0 \to 0$. Therefore, using Slutsky theorem,

$$\sqrt{n}(\hat{\theta_n} - \theta_0) \rightsquigarrow \mathcal{N}(0, (E\psi_{\theta_0}')^{-2}V_\psi).$$

Remark 8. We assume that the function $\theta \to \psi_\theta(X)$ possesses two continuous derivatives with respect to θ, for every X. This is not always true (e.g., the median). The condition can be replaced by a Lipschitz condition.